Kiosk Auto-Boot & VS Code AutoDebug Guide

Complete hardware-to-software replication steps for unattended Python debugging environments.

Hardware Prep: Beelink MiniPC Auto Power-On

Configure the Beelink MiniPC to automatically boot up as soon as it receives power.

1. Turn on the MiniPC and immediately tap the `Delete` key repeatedly until the BIOS setup screen appears.
2. Navigate to the **Advanced** or **Chipset** tab (menus vary slightly by specific Beelink model).
3. Look for one of the following settings:
   - **Chipset** > **PCH-IO Configuration** > **State After G3** > Set to `S0 State`
   - *OR* **Advanced** > **Power Management** > **Restore AC Power Loss** > Set to `Power On`
4. Press `F4` (or `F10` depending on the prompt at the bottom of the screen) to **Save & Exit**.

OS Prep: Windows 11 Autologon Setup

Configure Windows 11 to bypass the lock screen and automatically log in your specific user profile.

1. **Enable the Autologon Checkbox (often hidden in Win 11):**
   - Press `Win + R`, type `regedit`, and hit Enter.
   - Navigate to: `HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows NT\CurrentVersion\PasswordLess\Device`
   - Double-click `DevicePasswordLessBuildVersion` and change the Value data to `0`. Click OK and close Registry Editor.
2. **Configure Autologon:**
   - Press `Win + R`, type `netplwiz`, and hit Enter.
   - In the User Accounts window, **uncheck** the box that says *"Users must enter a user name and password to use this computer."*
   - Click **Apply**. A window will pop up asking for the user's password. Enter the password twice to confirm, then click OK.

Step 1: VS Code & File Migration

- Install **Visual Studio Code** and **Python** on the new machine.
- Copy your project workspace (e.g., `Gen2OpsConsole`) to the new machine. Make note of its new file path.

Step 2: Create the "AutoDebug" Profile

- Open VS Code on the new machine.
- Click the **Manage (Gear) Icon** in the bottom-left corner > **Profiles** > **Create Profile...**
- Name the profile `AutoDebug` and give it a distinctive icon/badge (like **AU**).
- VS Code will automatically switch to this new profile.

Step 3: Install Required Extensions

While the `AutoDebug` profile is active, open the Extensions tab and install:

- **Python** (by Microsoft) - Required for the debugger.
- **Auto Run Command** - The extension responsible for pressing F5 automatically on load.

Step 4: Configure the Profile's "Auto Run" Settings

Instruct the profile to start the debugger immediately after loading.

- Press `Ctrl + Shift + P`, type **Open Profile Settings (JSON)**, and hit Enter.
- Paste the following JSON configuration:

```
{
  "auto-run-command.rules": [
    {
      "command": "workbench.action.debug.start",
      "event": "onStartupFinished"
    }
  ],
  "workbench.startupEditor": "none"
}
```

Step 5: Define Launch Arguments (launch.json)

- Open your project folder/workspace in VS Code.
- Edit your `.vscode/launch.json` file to target your master script:

```
{
  "version": "0.2.0",
  "configurations": [
    {
      "name": "Gen2OpsConsole_Debug",
      "type": "debugpy",
      "request": "launch",
      "program": "${workspaceFolder}/MasterControl.py",
      "console": "integratedTerminal",
      "args": ["--arg1", "value1", "--arg2"]
    }
  ]
}
```

Step 6: Configure Windows Autostart (shell:startup)

Finally, set Windows to launch this specific workspace and profile automatically upon user logon.

- Press `Win + R`, type `shell:startup`, and press Enter.
- Right-click empty space in the folder > **New** > **Shortcut**.
- Paste your exact command line as the target (update the username/paths to match the new PC if different):

```
"C:\Users\NewUser\AppData\Local\Programs\Microsoft VS Code\Code.exe" "C:
\Users\NewUser\Documents\Software\Gen2OpsConsole\Gen2OpsConsole.code-workspace" --
new-window --profile "AutoDebug"
```